

Titanic Survival Data Analysis Report

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Tool Used: Python (Pandas, Matplotlib, Seaborn)

Objective: Extract insights using visual and statistical exploration.

The main goal of this project is to perform an analysis of the Titanic dataset to uncover meaningful insights about the passengers and the factors that influenced their chances of survival. The tool I used for this project is Python and its libraries Pandas for data manipulation, Matplotlib and Seaborn for visualization.

Dataset Overview

The dataset contains details of passengers including demographic data, ticket details, and whether they survived or not. It includes both categorical and numerical features, useful for predictive analysis and insight generation.

Data dimensions:

- Total Rows: 891
- Total Columns: 12

Main columns:

- Survived – Survival (0 = No, 1 = Yes)
- Pclass – Passenger class (1 = 1st, 2 = 2nd, 3 = 3rd)
- Sex, Age, SibSp, Parch, Fare, Embarked

Data Summary

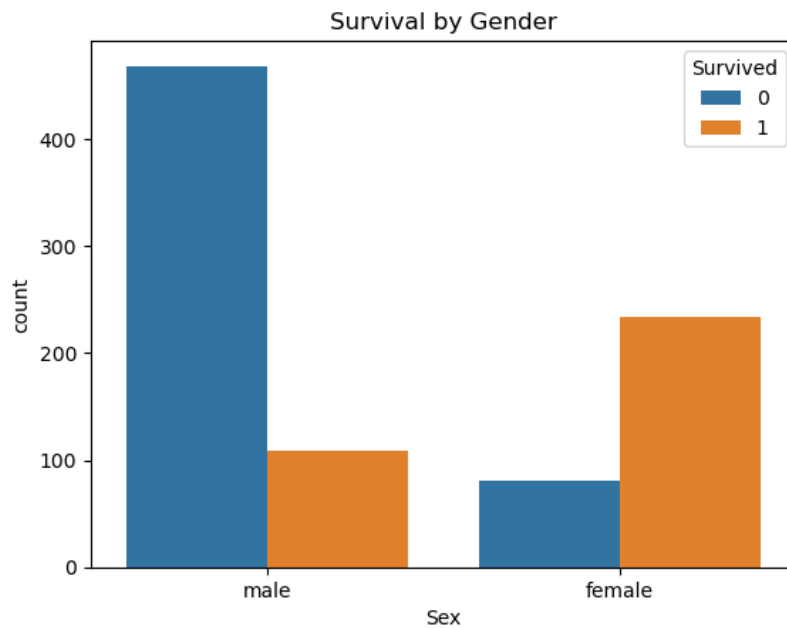
Total entries: 891

Missing values found and handling methods used:

- **Age:** 177 missing. Filled with median age
- **Cabin:** 687 missing.
- **Embarked:** 2 missing. Filled with most frequent port (S)

Data Visualization

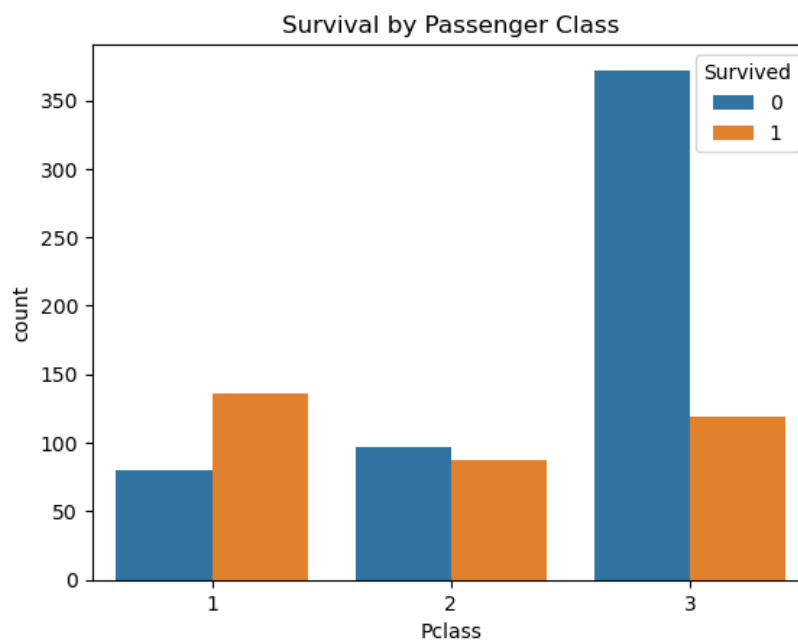
1. Survival by Gender



Observations:

- Most males did not survive.
- Females had a high survival rate

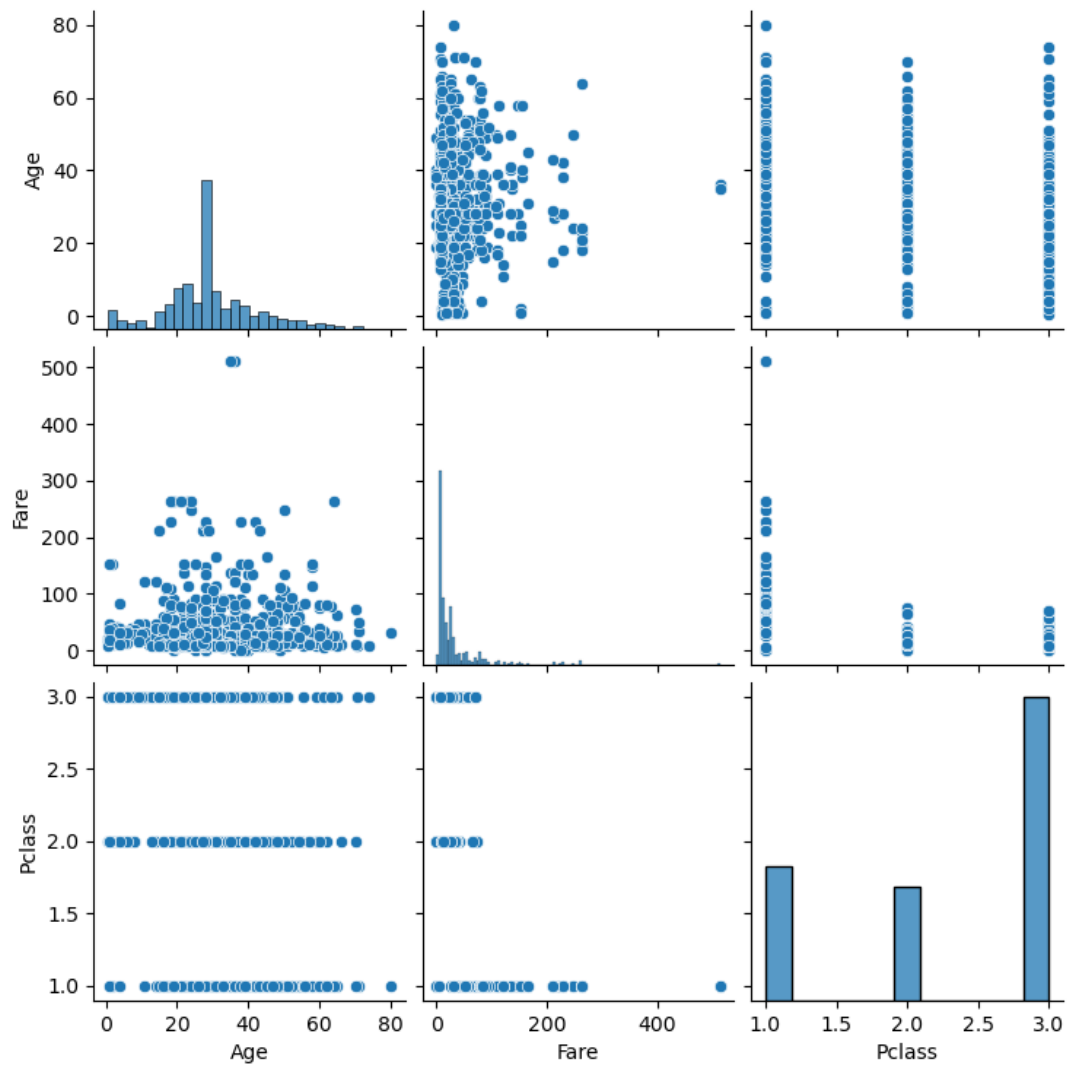
2. Survival by Class



Observations:

- Higher survival rates in 1st passenger class

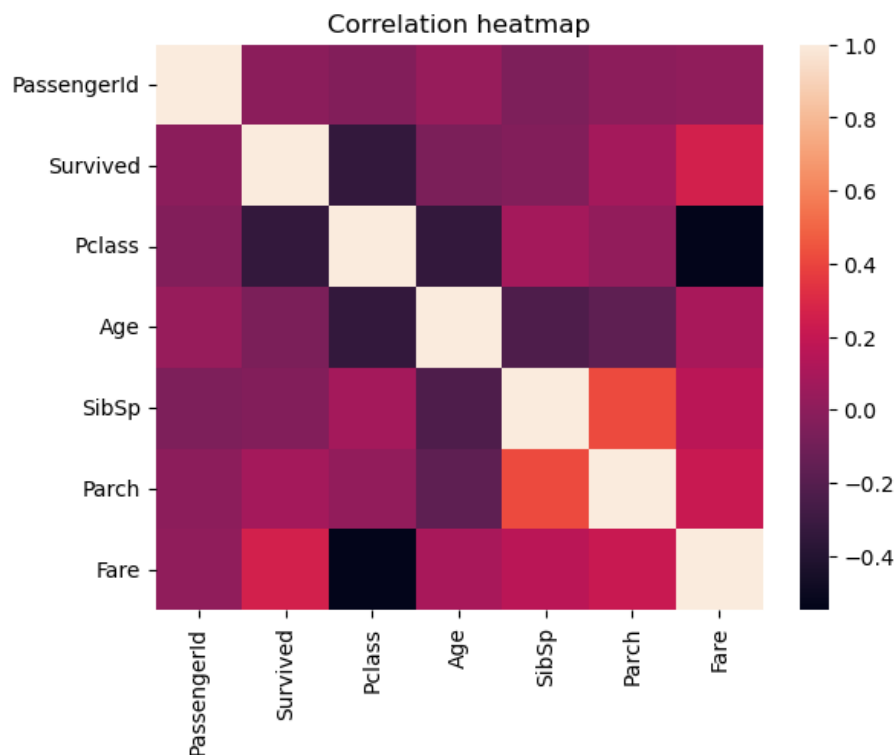
3. Pairwise Correlation



Observations:

- Higher fares are loosely correlated with survival.
- Most survivors are from younger age ranges and higher classes.

4. Correlation Heatmap



Observations:

- Fare and Pclass are negatively correlated.
- Survived has slight correlation with Fare and Pclass.

Key Insights

- Gender is a strong factor here in survival, Females were prioritized for survival.
- Class mattered, Passengers in 1st class had better chances since they too were given high priority.
- Fare may reflect social status, showing a mild correlation with survival.
- Age did not show a strong trend but extreme age groups (children) were more likely to survive.

Conclusion

Our statistical and visual analysis of the data revealed some distinct trends, particularly in relation to fare, passenger class, and gender. These patterns closely reflect what we know from history that is, people from higher classes and women had a better chance of survival. this shows how social and economic status played a major role during the Titanic disaster.